



Grundlagen der IT

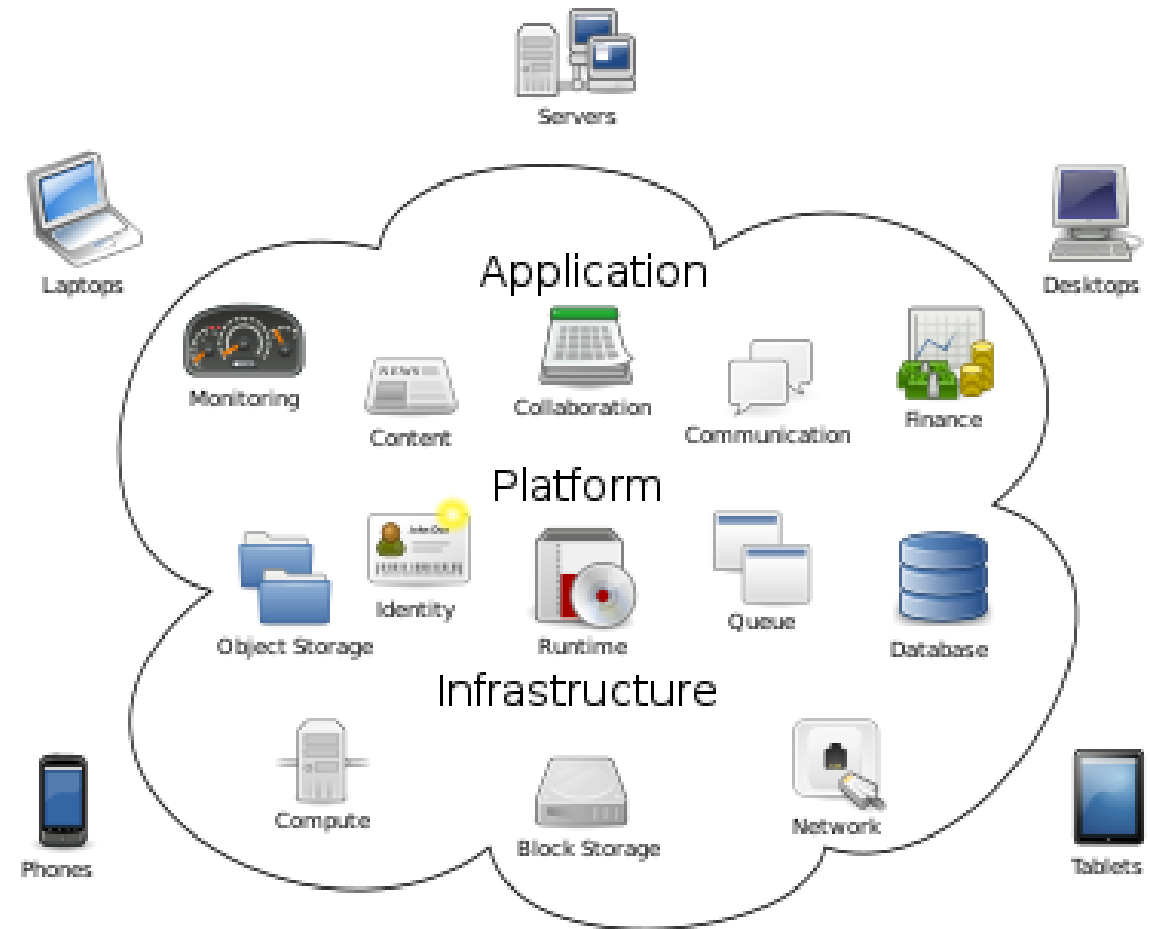
Exkurs: Cloud Computing

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Cloud Computing

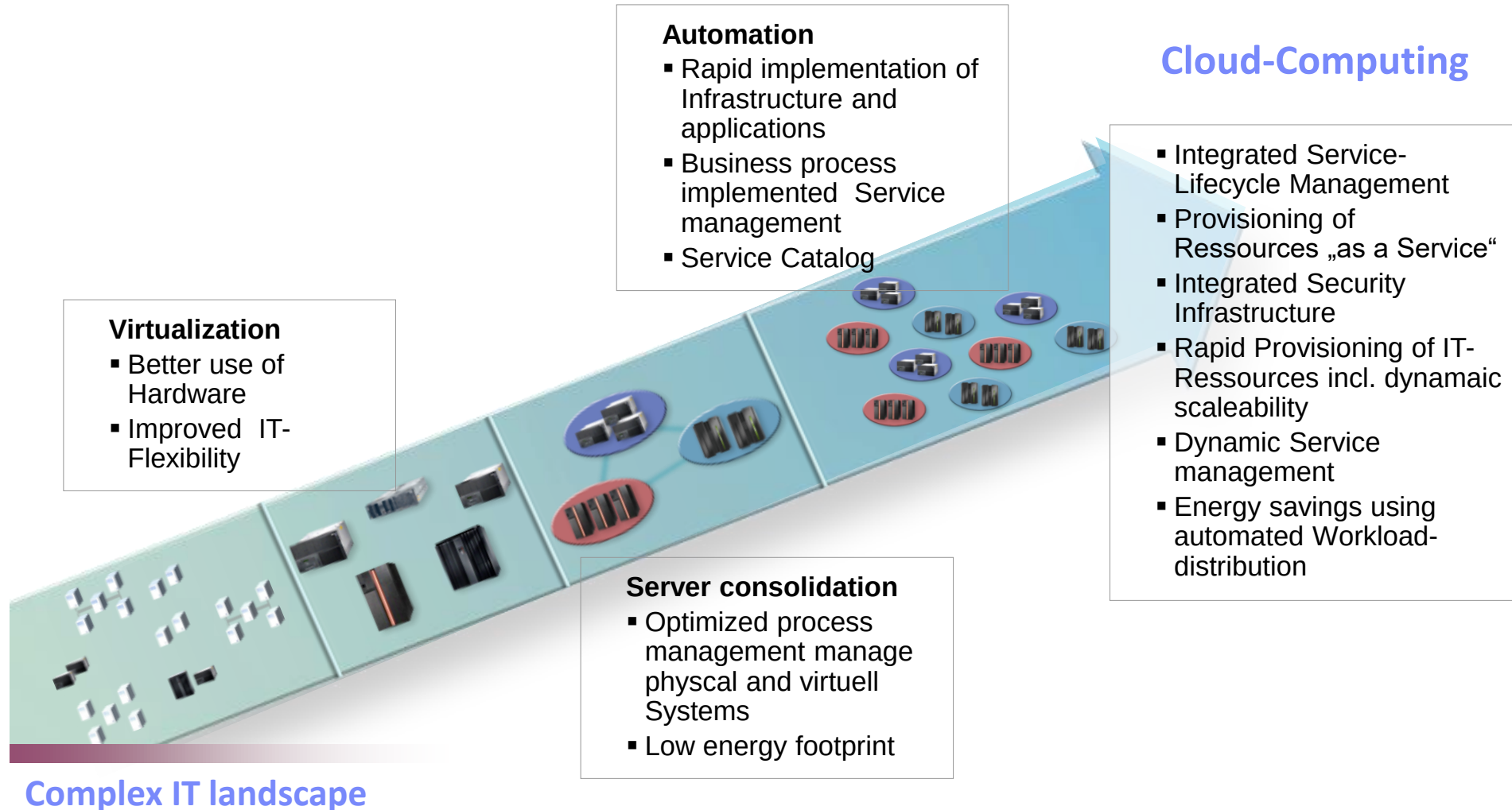
- Cloud computing is the on-demand availability of computer system resources, especially storage, computing power and network including applications, without direct active management by the user.
- Large clouds often have functions distributed over multiple locations, each of which is a data center. Cloud computing relies on sharing of resources to achieve coherence and typically uses a **pay-as-you-go business model**, which can help in reducing capital expenses but may also lead to unexpected operating expenses for users.
- Origin: Strategic Outsourcing, Grid Computing and Amazon
- First vendors: Amazon, Microsoft, IBM, Google, Oracle and many other providers
- According to International Data Corporation (IDC), global spending on cloud computing services has reached \$706 billion and expected to reach \$1.3 trillion by 2025. While Gartner estimated that global public cloud services end-user spending would reach \$600 billion by 2023.



Cloud Computing, Definition

- The United States National Institute of Standards and Technology's definition of cloud computing identifies "five essential characteristics":
 - **On-demand self-service.** A consumer can provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction.
 - **Broad network access.** Capabilities are available over the network and accessed through standard mechanisms that enable use by thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).
 - **Resource pooling.** The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
 - **Rapid elasticity.** Capabilities can be elastically provisioned and released, in some cases automatically. To the consumer, the capabilities available for provisioning often appear unlimited and can be appropriated in any quantity at any time.
 - **Measured service.** Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

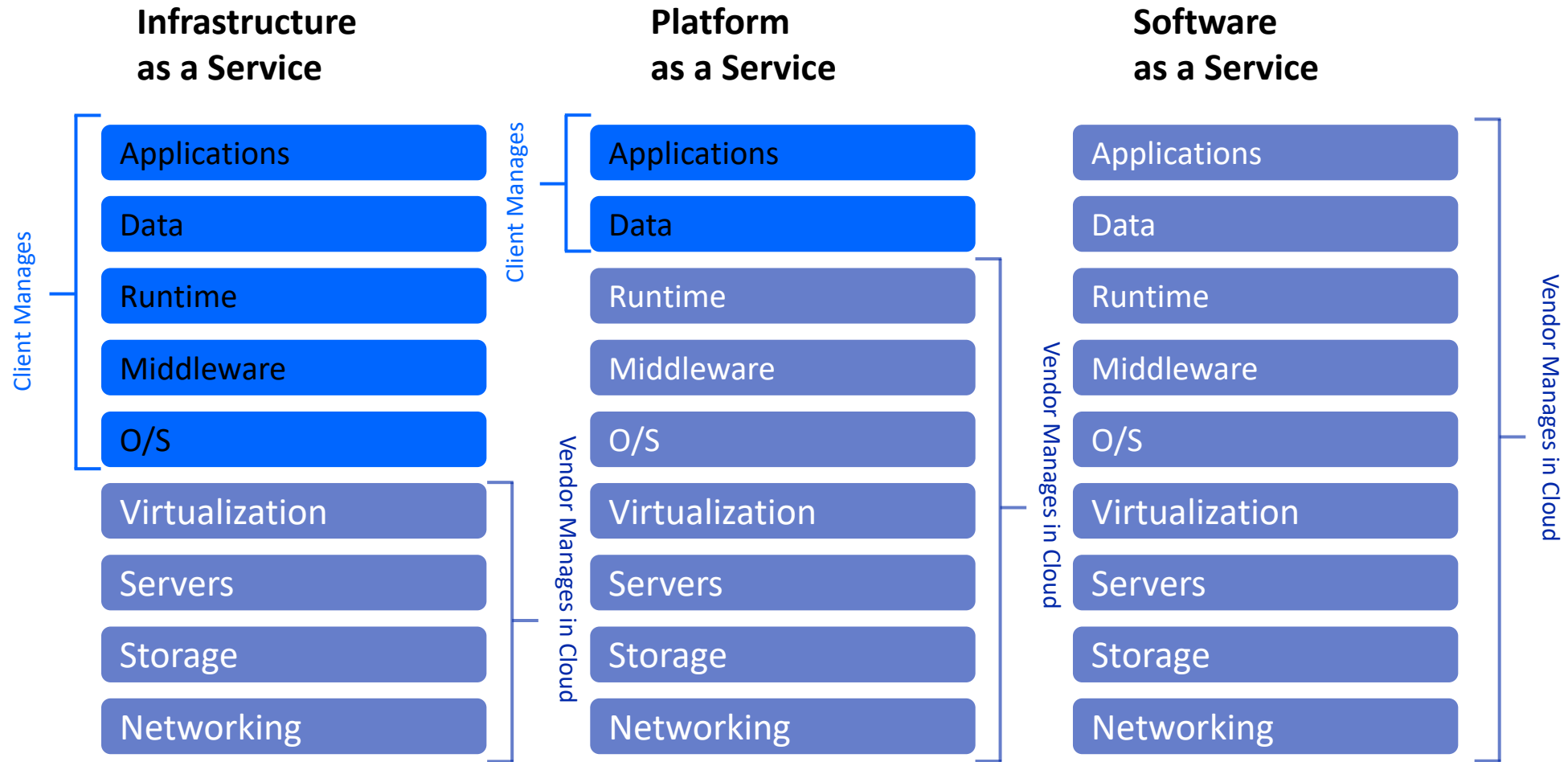
Technology Evolution towards Cloud Computing



Cloud Computing, Value Proposition

- Cost reduction?
- Device Independence: Device and location independence enable users to access systems using a web browser regardless of their location
- Maintenance: Maintenance of cloud environment is easier because the data is hosted on an outside server maintained by a provider without the need to invest in data center hardware.
- Multitenancy: Multitenancy enables sharing of resources and costs across a large pool of users.
- Performance: Performance is monitored by IT experts from the service provider.
- Productivity
- Availability: Provided by Cloud provider
- Scalability and Elasticity: Scalability and elasticity via dynamic ("on-demand") provisioning of resources on a fine-grained, self-service basis.
- Security is a big issue

Cloud Service Models



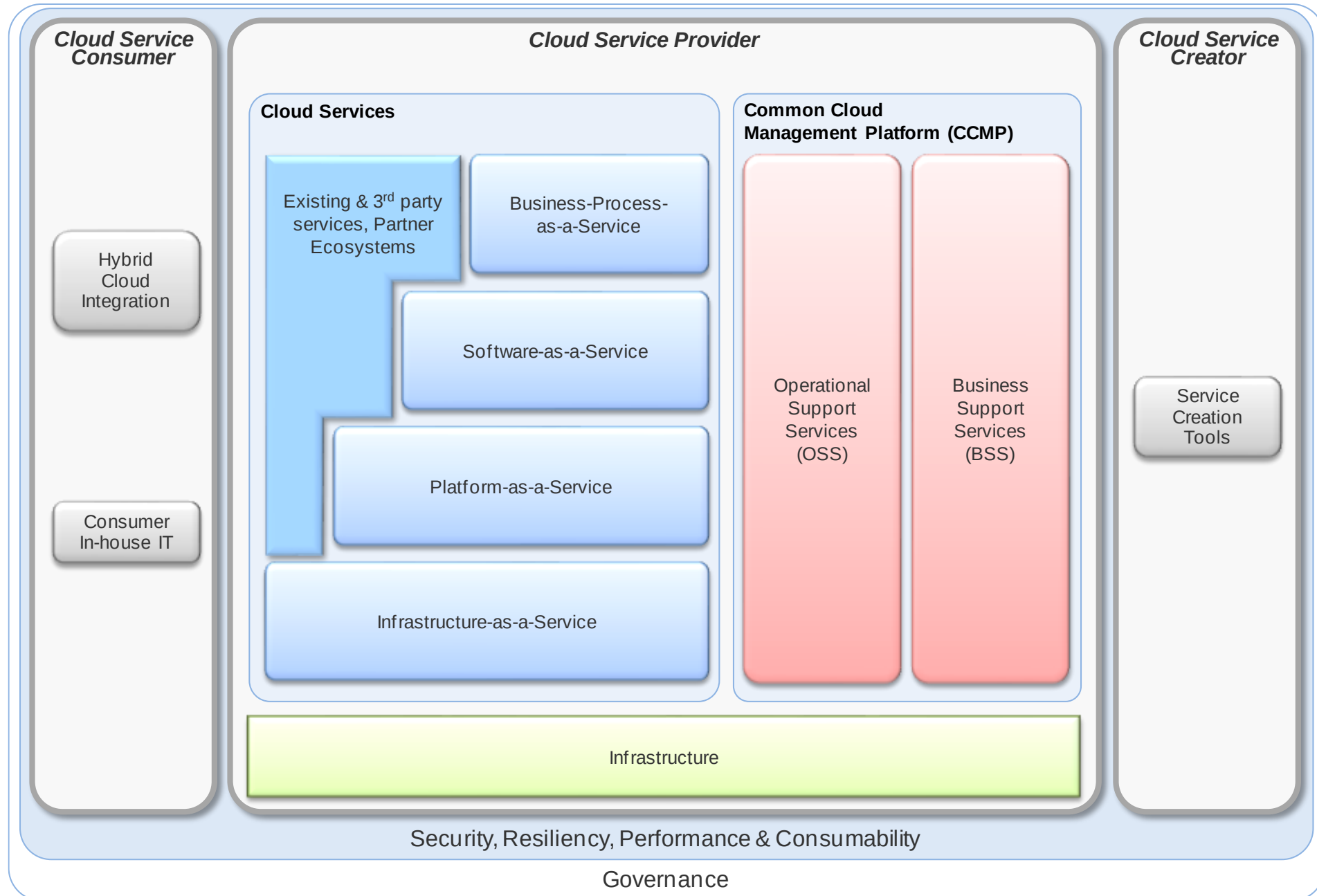
Customization; higher costs; slower time to value

Standardization; lower costs; faster time to value

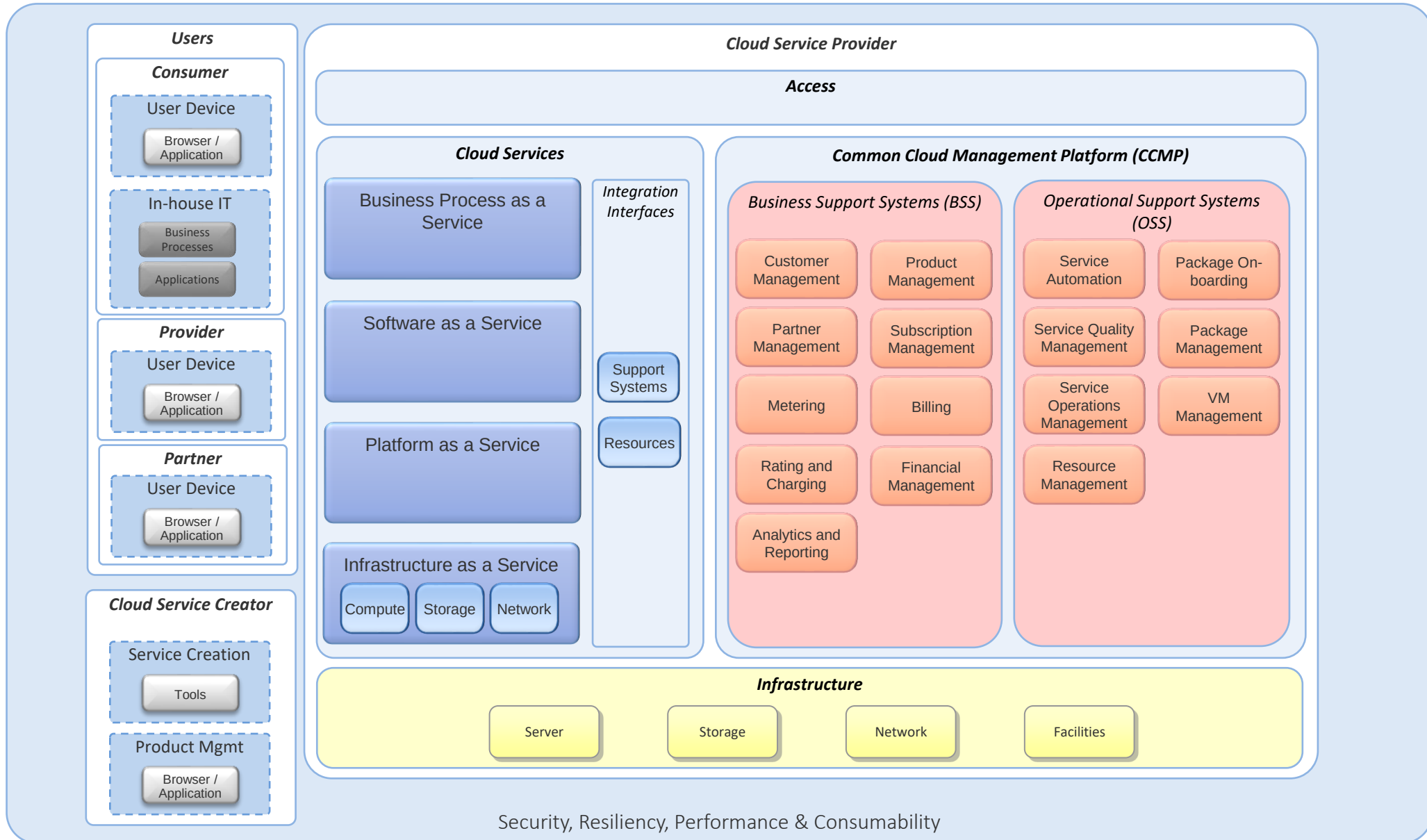
New Cloud Computing Service Models

- Function as a Service: Function as a service (FaaS) is a service-hosted remote procedure call that utilizes serverless computing to enable deploying individual functions in the cloud to run in response to events. Requests are billed by an abstract measure of the resources required to satisfy the request, rather than per virtual machine per hour.
- Examples:
 - Financial Services
 - Weather Services
 - Google Maps and others
 - High computing resources (HPC)
 - Big Data
 - ...

Cloud Computing Architecture



Cloud Computing Architecture – Drill-down



Cloud Computing, Deployment Models

Private

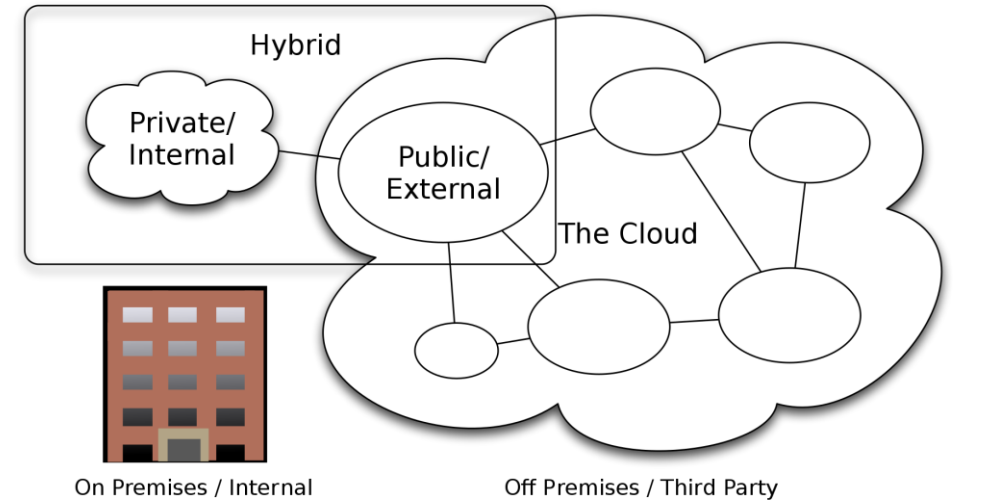
Private cloud is cloud infrastructure operated solely for a single organization, whether managed internally or by a third party, and hosted either internally or externally

Public

Cloud services are considered "public" when they are delivered over the public Internet, and they may be offered as a paid subscription, or free of charge

Hybrid

Hybrid cloud is a composition of a public cloud and a private environment, such as a private cloud or on-premises resources,[68][69] that remain distinct entities but are bound together, offering the benefits of multiple deployment models.



Cloud Computing Types

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Cloud Computing, Potential Issues

- Location of Cloud compute center, based on:
 - Customer's preferences and strategy
 - Industry specific issues
 - Legal issues
- Security questions related to Cloud management locations
- Cloud-in and Cloud-out
- Data transfer into the Cloud, security?
- Cloud maintenance intervals, high availability in the Cloud
- Pricing of Cloud providers
 - Contract with 'full' dynamic requests
 - Contract for business applications, longer term period